EKAM MANN

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Open to Relocation

EDUCATION

The University of British Columbia — Vancouver, BC

Bachelor of Applied Science in Mechanical Engineering — Expected May 2029

- 3.8 GPA | Dean's List | RCAF Foundation Scholar

TECHNICAL SKILLS

- **CAD and Simulation:** SolidWorks (Surfacing, FEA, CAD-to-CAM), ANSYS CFD, AutoCAD, CAEPIPE
- **Manufacturing:** CNC Machining, Manual Lathe and Mill, Waterjet Cutting, Vacuum Thermoforming, Composite Layup, FDM 3D Printing
- **Process Methods:** DFM/DFA, FMEA, Tolerance Analysis, GD&T, Lean Principles (5S, Waste Reduction)
- **Codes and Standards:** ASME Y14.5, ASME B31.3, ASME B16.5, CSA Z662
- **Programming and Controls:** MATLAB, Python, Arduino (C/C++), MS Excel

EXPERIENCE

Petra Engineers and Consultants Ltd. — Calgary, AB

Mechanical Design Engineering Intern — May 2026 - August 2026

- Flagged 6 pipe support locations for senior engineering review, by performing pipe stress analysis in CAEPIPE for a natural gas distribution system and interpreting code stress results against allowables.
- Documented 11 review comments incorporated into issued revisions, by redlining 4 engineering drawing packages (P&IDs, site plans, demolition plans) against ASME B31.3, CSA Z662, and ASME B16.5 requirements.
- Produced wall thickness summaries used in project reports and construction drawings, by calculating minimum required thickness for 12 line segments with a CSA Z662 design spreadsheet under senior engineer supervision.
- Corrected 14 discrepancies across 87 reviewed line items, by verifying bills of materials, equipment lists, and valve lists against AutoCAD isometric drawings for a regulator station project scope.
- Enabled as-built record updates for an operating oil and gas facility, by documenting 7 missing or illegible equipment tags during site visits and producing structured field deviation logs.

UBC Solar — Vancouver, BC

Aeroshell Designer — September 2025 - Present

- Improved aeroshell aerodynamic performance (Cd: 0.176 → 0.110, Cl: -1.0 → -0.55) through iterative geometry redesign, validated using steady RANS (k-ω SST) CFD simulations with refined near-wall mesh resolution.
- Contributed to 3 of 7 aeroshell design iterations toward a manufacturing-ready release by modelling topshell and canopy geometry in SolidWorks surfacing, incorporating iterative design feedback and maintaining revision-controlled CAD outputs.
- Accelerated onboarding of 12 team members in solar car manufacturing by developing 4 standardized process documents and 2 training videos covering composite layup and thermoforming procedures.

ENGINEERING PROJECTS

Polycarbonate Windshield: Design, Thermoforming, and Failure Analysis — *UBC Solar*

- Achieved approximately 2.4% drag reduction over baseline per CFD, by modelling a double-curvature polycarbonate windshield in SolidWorks to improve laminar flow transition between canopy and aeroshell.
- Cut estimated part cost from $2,000 (machined polycarbonate) to $300, by developing a vacuum thermoforming process and using advanced surfacing techniques to design a waterjet-cut thermoforming jig intended for use in an industrial oven.
- Eliminated a stress concentration at the adhesive bondline under peak thermal loading, by conducting FEA-based failure analysis and redesigning the CFRP L-flange attachment geometry.
- Selected the polycarbonate-to-CFRP bonded joint system, by evaluating Sikaflex 252 against MS polymer, VHB tape, and neutral-cure silicone alternatives for thermal expansion compatibility and serviceability.

Vortex Generator Research: NACA 2412 Stall Characteristics — *Grant-Funded Independent Research*

- Improved stall angle of a NACA 2412 airfoil by 76%, by designing and wind tunnel testing 14 vortex generator configurations.
- Enabled all experimental runs for under $50, by designing and building a self-constructed wind tunnel from scratch.